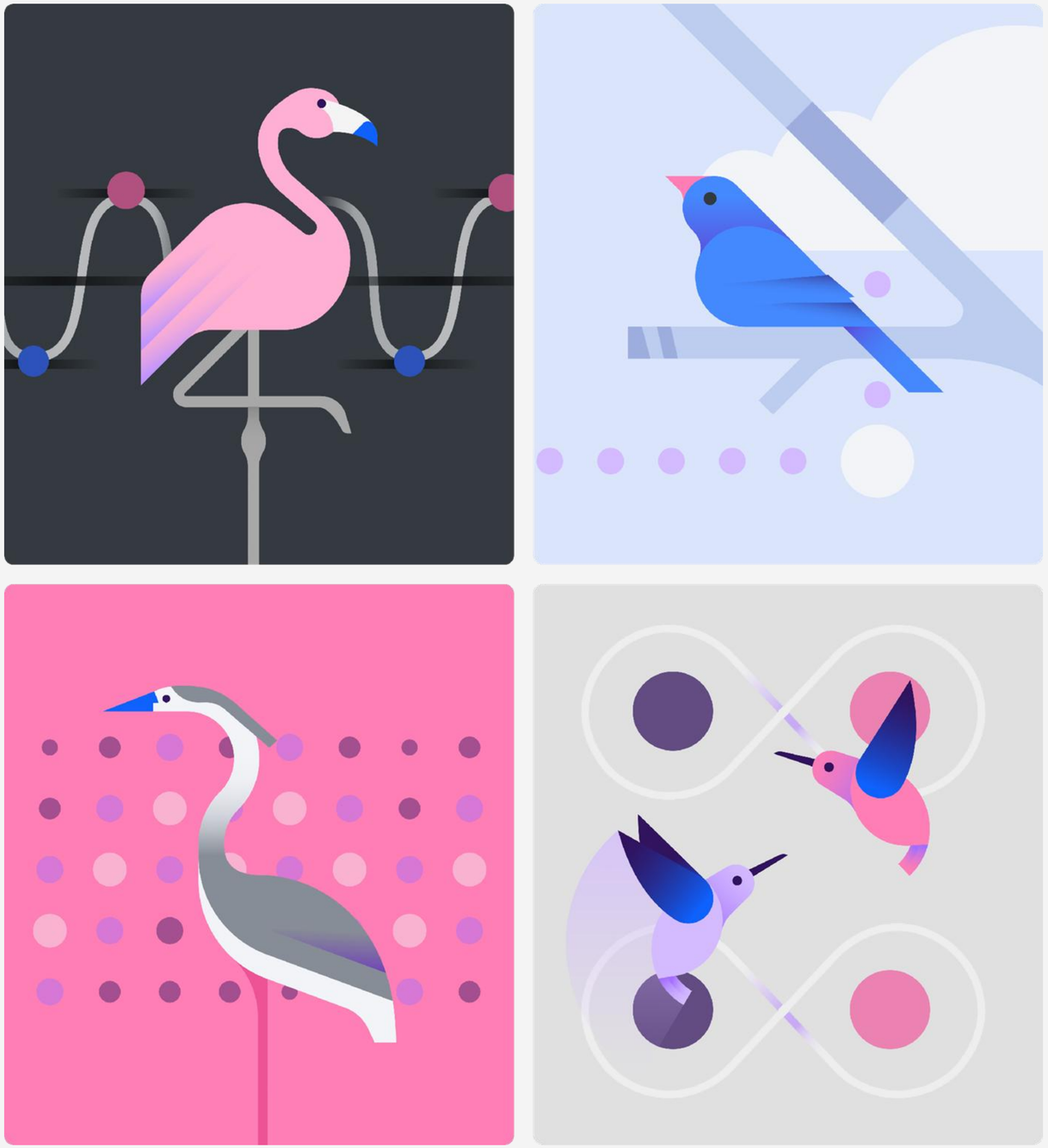
Qiskit Fall Fest 2026

Quantum Hardware

Dr. Zuhair Ahmed
Founder & Chairman, CETQAP
Quantum Computing
Researcher & Educator



The building blocks of quantum computers

Particles Can Communicate Through Entanglement Without Sending Messages: Two entangled particles can display correlated measurement outcomes even when separated by enormous distances. What's surprising is that these correlations cannot be explained by ordinary classical physics, although they cannot be used to send information faster than light.



Quantum computers are built on several fundamental principles that make them different from classical computers. Their basic units of information are called qubits (quantum bits), which use quantum properties such as superposition and entanglement to represent and process information. Unlike classical bits, which can only be 0 or 1, qubits can exist in a combination of both states, allowing quantum computers to perform complex computations in fundamentally different ways.

The key building blocks of quantum computers include qubits, superposition, entanglement, quantum gates, quantum circuits, measurement, quantum hardware, and quantum error correction. Together, these components enable quantum computers to execute specialized algorithms with potential applications in artificial intelligence, cryptography, drug discovery, optimization, and scientific research.

What makes quantum hardware different

A classical computer stores information as bits, each either a 0 or a 1.
A quantum computer uses qubits instead.

Thanks to superposition, a qubit can represent 0 and 1 at the same time, and many qubits together can hold far more information than the same number of bits.

Qubits can also become entangled, so the state of one instantly affects another, even far apart.

Quantum hardware is the physical machinery that creates, controls and protects qubits long enough to run a calculation — chips, lasers, magnets and extreme cooling systems working together.

Your Body Is Full of Quantum Phenomena: Every atom in your body follows the laws of quantum mechanics. The chemistry that allows your cells, DNA, and biological molecules to function depends on quantum behavior.

No qubits, no quantum computer



IBM’s Condor chip has passed 1,000 superconducting qubits on a single chip[1]

↗ 1,000+

Best gate fidelities on leading hardware now exceed 99.9% per operation [2]

99.9%

Source: 1. IBM Quantum roadmap, 2025
Source: 2. IBM & Google Quantum AI published benchmarks

Two ways to build a qubit

Superconducting qubits

Tiny circuits cooled to near absolute zero, where electric current flows without resistance.

Used by IBM and Google; fast and easy to manufacture with chip-making techniques.

Needs dilution refrigerators colder than deep space, and qubits lose their state quickly.

Trapped-ion qubits

Individual charged atoms held in place and controlled with laser beams.

Used by IonQ and Quantinuum; very stable with long coherence times and high accuracy.

Operations run slower, which makes scaling up to many qubits more difficult.



Difference Between Quantum Computer & Quantum Simulation



A Quantum Computer is a machine that uses quantum bits (qubits) instead of traditional computer bits to perform calculations. It uses principles such as superposition and entanglement to solve certain complex problems in ways that classical computers may struggle with. Think of it as a new type of computer that processes information using the laws of quantum physics. Quantum Simulation is the process of recreating and studying the behavior of quantum systems using computational models.
|
It allows scientists and researchers to experiment with quantum particles, molecules, chemical reactions, and quantum circuits without needing to build expensive physical quantum hardware. Why Is Quantum Simulation So Important? Quantum simulation is extremely important because building and maintaining actual quantum computers is incredibly expensive, technically challenging, and requires highly controlled environments. Real quantum hardware is also affected by noise, errors, and limited qubit availability. Simulators allow researchers, students, and developers to experiment, test algorithms, understand quantum behaviour, and develop new technologies at a much lower cost. They can also simulate ideal quantum systems without the hardware noise that affects real devices.

A quantum computer is the actual machine, while quantum simulation is the virtual laboratory where we can design, test, experiment, and understand quantum computing. In fact, simulation is essential for developing quantum technology because we need to understand, test, and improve quantum systems before relying on physical hardware.

Four open problems hardware teams are racing to solve

1

Decoherence. Qubits lose their fragile quantum state within microseconds, from heat, vibration or stray electromagnetic noise.

2

Error correction. Many physical qubits must team up to form one reliable "logical" qubit, since individual qubits are error-prone.

3

Extreme cooling. Superconducting qubits need refrigerators near absolute zero, which are large, costly and hard to scale.

4

Scalability. Connecting thousands of qubits while keeping them stable and well controlled remains a major engineering hurdle.

Why it matters

What quantum hardware could unlock

Cryptography

Breaking and building the next generation of secure encryption.

Drug discovery

Simulating molecules accurately to speed up finding new medicines.

Optimization

Solving complex logistics, finance and scheduling problems faster.

Materials science

Designing new materials, batteries and catalysts atom by atom.



Facts People Know less about Quantum Computing

Have you ever wondered what connects your consciousness, your body, and the entire universe? Everything around us, from distant galaxies to the cells inside your body, is made of atoms. But atoms themselves are not tiny solid balls. They consist of a nucleus surrounded by electrons, and their behavior is governed by quantum mechanics. At an even deeper level, the particles that make up atoms are described through quantum fields, which are fundamental to our modern understanding of nature. In a fascinating sense, the same laws of quantum physics that govern matter throughout the universe also govern the building blocks of our bodies.

the atoms that make up your body were forged through cosmic history. Many elements were created inside stars and distributed throughout space through stellar explosions and other cosmic events. Eventually, those elements became part of planets, oceans, living organisms, and ultimately you. In this way, we are physically connected to the history of the universe. Quantum mechanics helps explain the behavior of the matter that makes this possible, while consciousness raises a deeper question: How does a universe made of physical matter give rise to beings capable of understanding the universe itself?

Facts People Know less about Quantum Computing

The Sun Would Struggle to Shine Without Quantum Tunneling: The Sun's nuclear fusion reactions depend partly on particles tunneling through energy barriers that classical physics says they cannot cross with their available energy.

Quantum Particles Have No Definite Classical Path: Before measurement, quantum systems can be described by probability amplitudes rather than a single definite trajectory. This means the familiar idea of a tiny particle following a predictable path does not always apply.

Time Can Flow Differently for Different Observers: Atomic clocks can detect differences in the passage of time caused by gravitational fields and motion. Quantum technology makes these incredibly precise measurements possible.

A Quantum Particle Can Have Negative Energy Density: Under special conditions, quantum field theory permits regions with negative energy density relative to the vacuum. Scientists have studied such effects in phenomena such as the Casimir effect.

Quantum Mechanics Makes It Possible to See Individual Atoms: Modern scanning tunnelling microscopes use quantum tunnelling currents to map surfaces at atomic scales. Scientists can even manipulate individual atoms using specialized instruments.

The Universe May Be More Connected Than It Appears: Quantum entanglement reveals that nature cannot always be described as separate objects possessing independent properties. Experiments testing Bell's inequalities have confirmed correlations that challenge our everyday understanding of physical reality.

The road ahead

Quantum hardware is still young, but progress is fast: qubit counts, stability and error correction are all improving year over year.

Superconducting qubits, trapped ions and other hardware are converging toward that goal from different directions.

The goal is a fault-tolerant quantum computer, one that can run long, complex programs without errors piling up.

Dr. Zuhair Ahmed — Founder & Chairman, CETQAP.

Linkedin: /in/zytabyte
Instagram: drzuhairahmed
Youtube: drzuhairahmed



QISKIT 2026 2026 FALL FEST